

Take Home Activity 06

Student Number – 35040

Q1

```
public class Device {  
    String brand;  
    String model;  
    boolean isOn;  
  
    public void turnOn() {  
        isOn = true;  
        System.out.println(brand + model + " is now On");  
    }  
  
    public void turnOff() {  
        isOn = false;  
        System.out.println(brand + model + " is now Off");  
    }  
}
```

```
public class SmartDevice extends Device{  
    String appName;  
  
    public void connectToApp(){  
        System.out.println(brand + " Connected to " + appName);  
    }  
}
```

```
class Main {  
    Run | Debug  
    public static void main(String[] args) {  
        SmartDevice s1 = new SmartDevice();  
        s1.brand = "Apple";  
        s1.model = "Iphone 18 pro max";  
        s1.appName = "Smart Things";  
        s1.turnOn();  
        s1.connectToAPp();  
        s1.turnOff();  
    }  
}
```

```
AppleIphone 18 pro max is now On  
Apple Connected to Smart Things  
AppleIphone 18 pro max is now Off
```

Q2

```
package Q2;
public class NetworkDevice {
    String ipAddress;
    String macAddress;

    public void connect() {
        System.out.println("Connecting device with IP: " + ipAddress);
    }

    public void disconnect() {
        System.out.println("Device " + ipAddress + " disconnected");
    }
}
```

```
package Q2;
public class Firewall extends NetworkDevice{
    String firewallName;

    public void blockTraffic(String source) {
        System.out.println(firewallName + "blocked traffic from" + source);
    }

    public void allowTraffic(String source) {
        System.out.println(firewallName + "allowed traffic from" + source);
    }
}
```

```

package Q2;

public class Main {
    Run | Debug
    public static void main(String []args) {
        Firewall f1 = new Firewall();
        f1.ipAddress = "192.168.1.1";
        f1.macAddress = "AA:BB:CC:DD:EE:FF";
        f1.firewallName = "PFSense";
        f1.connect();
        f1.blockTraffic(source: "10.0.0.5");
        f1.allowTraffic(source: "10.0.0.9");
        f1.disconnect();
    }
}

```

```

in'
Connecting device with IP: 192.168.1.1
PFSenseblocked traffic from10.0.0.5
PFSenseallowed traffic from10.0.0.9
Device 192.168.1.1 disconnected

```

Q3

```
package Q3;

public class Person {
    String name;
    int age;

    public void introduce() {
        System.out.println("Hi, I'am " + name + " and I am " + age + " years old.");
    }
}
```

```
package Q3;

public class Employee extends Person {
    String employeeId;
    String department;

    public void showEmployeeInfo() {
        System.out.println("Employee " + employeeId + " works in " + department);
    }
}
```

```
package Q3;

public class SecurityAnalyst extends Employee{
    int clearanceLevel;

    public void analyzeThreats() {
        System.out.println("Analyst " + name + " is analysing threats at clearance level " + clearanceLevel);
    }
}
```

```

package Q3;

public class Main {
    Run | Debug
    public static void main(String[] args) {
        SecurityAnalyst s1 = new SecurityAnalyst();
        s1.name = "Diana";
        s1.age = 29;
        s1.employeeId = "SA-201";
        s1.department = "SOC";
        s1.introduce();
        s1.showEmployeeInfo();
        s1.analyzeThreats();
    }
}

```

Hi, I'am Diana and I am 29 years old.

Employee SA-201 works in SOC

Analyst Diana is analysing threats at clearance level 0

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Q4

```

package Q4;

public class Vehicle {
    String make;
    int speed;

    public void move() {
        System.out.println( make + " is moving at" + speed + " km/h");
    }
}

```

```
package Q4;

public class ElectricVehicle extends Vehicle {
    int batteryLevel;

    public void chargeBattery() {
        batteryLevel = 100;
        System.out.println(make + " battery fully charged.");
    }

    public void showBattery() {
        System.out.println("Battery: " + batteryLevel + "");
    }
}
```

```
package Q4;

public class AutonomousVehicle extends ElectricVehicle {
    boolean autopilotEnabled;

    public void enableAutopilot() {
        System.out.println("Autopilot ON for " + make);
    }
}
```

```

package Q4;

public class Main {
    Run | Debug
    public static void main(String []args) {
        AutonomousVehicle a1 = new AutonomousVehicle();
        a1.make = "Tesla";
        a1.speed = 120;
        a1.batteryLevel = 45;
        a1.move();
        a1.showBattery();
        a1.chargeBattery();
        a1.enableAutopilot();
    }
}

```

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Tesla is moving at 120 km/h

Battery: 45

Tesla battery fully charged.

Autopilot ON for Tesla

Q5


```
package Q5;

public class Account {
    String accountNumber;
    String holderName;
    double balance;

    public void showDetails() {
        System.out.println("Account " + accountNumber + " | Holder: " + holderName + " | Balance: " + balance);
    }

    public void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited " + amount + " New balance: " + balance);
    }
}
```

```
package Q5;

public class SavingsAccount extends Account {
    double interestRate;

    public void applyInterest() {
        balance += (balance * interestRate / 100);
        System.out.println("Interest applied. New balance: " + balance);
    }
}
```

```
package Q5;

public class LoanAccount extends Account {
    double loanAmount;

    public void repayLoan(double amount) {
        loanAmount -= amount;
        System.out.println("Loan repaid: " + amount + " Remaining: " + loanAmount);
    }
}
```

```
package Q5;

public class FixedDepositAccount extends Account {
    String maturityDate;

    public void showMaturity() {
        System.out.println("Fixed deposit matures on: " + maturityDate);
    }
}
```

```
package Q5;
```

```
public class Main {
```

```
    Run | Debug
```

```
    public static void main(String[] args) {
```

```
        SavingsAccount savings = new SavingsAccount();
```

```
        savings.accountNumber = "SA001";
```

```
        savings.holderName = "Navitha";
```

```
        savings.balance = 10000;
```

```
        savings.interestRate = 5;
```

```
        savings.showDetails();
```

```
        savings.applyInterest();
```

```
        System.out.println();
```

```
        LoanAccount loan = new LoanAccount();
```

```
        loan.accountNumber = "LA001";
```

```
        loan.holderName = "Navitha";
```

```
        loan.balance = 0;
```

```
        loan.loanAmount = 50000;
```

```
        loan.showDetails();
```

```
        loan.repayLoan(amount: 10000);
```

```
        System.out.println();
```

```
        FixedDepositAccount fixed = new FixedDepositAccount();
```

```
        fixed.accountNumber = "FD001";
```

```
        fixed.holderName = "Navitha";
```

```
        fixed.balance = 200000;
```

```
        fixed.maturityDate = "2026-12-31";
```

```
        fixed.showDetails();
```

```
        fixed.showMaturity();
```

```
    }
```

```
}
```

```
a:\jdk_ws\Take Home Activity 00_79191555\bin> Q5.Main
Account SA001 |Holder: Navitha | Balance: 10000.0
Interest applied. New balance: 10500.0

Account LA001 |Holder: Navitha | Balance: 0.0
Loan repaid: 10000.0 Remaining: 40000.0

Account FD001 |Holder: Navitha | Balance: 200000.0
Fixed deposit matures on: 2026-12-31
```

Q6

```
package Q6;

class Threat {
    String threatId;
    String source;
    String severity;

    public void logThreat() {
        System.out.println("[THREAT LOG] ID: " + threatId + " | Source: " + source + " | Severity:" + severity);
    }

    public void raiseAlert() {
        System.out.println("!! SECURITY ALERT !! Threat " + threatId + " detected from " + source);
    }
}
```

```
package Q6;

public class MalwareAttack extends Threat{
    String malwareType;

    public void quarantine() {
        System.out.println("Quarantining " + malwareType + " from " + source);
    }
}
```

```
package Q6;
```

```
public class PhishingAttack extends Thread{
```

```
    String targetEmail;
```

```
    public void blockEmail() {
```

```
        System.out.println("Blocking phishing email targeting: " + targetEmail);
```

```
    }
```

```
}
```

```
package Q6;
```

```
public class DDoSAttack extends Thread {
```

```
    int packetCount;
```

```
    public void mitigate() {
```

```
        System.out.println("Mitigating DDoS from " + source + " -- " + packetCount + " packets dropped");
```

```
    }
```

```
}
```

Main.java 7 () Q6

package Q6;

public class Main {

Run | Debug

public static void main(String[] args) {

 MalwareAttack malware = new MalwareAttack();

 malware.threatId = "T001";

 malware.source = "192.168.1.10";

 malware.severity = "Critical";

 malware.malwareType = "Ransomware";

 malware.logThreat();

 malware.raiseAlert();

 malware.quarantine();

 System.out.println();

 PhishingAttack phishing = new PhishingAttack();

 phishing.threatId = "T002";

 phishing.source = "fake-login@email.com";

 phishing.severity = "High";

 phishing.targetEmail = "student@nsbm.ac.lk";

 phishing.logThreat();

 phishing.raiseAlert();

 phishing.blockEmail();

 System.out.println();

 DDoSAttack ddos = new DDoSAttack();

 ddos.threatId = "T003";

 ddos.source = "203.94.55.12";

 ddos.severity = "Medium";

 ddos.packetCount = 5000;

 ddos.logThreat();

 ddos.raiseAlert();

 ddos.mitigate();

}

}

```
a\jdt_ws\Take Home Activity 06_79f9f533\bin' 'Q6.Main'
```

```
[THREAT LOG] ID: T001 | Source: 192.168.1.10 | Severity:Critical
```

```
!! SECURITY ALERT !! Threat T001 detected from 192.168.1.10
```

```
Quarantining Ransomware from 192.168.1.10
```

```
[THREAT LOG] ID: T002 | Source: fake-login@email.com | Severity:High
```

```
!! SECURITY ALERT !! Threat T002 detected from fake-login@email.com
```

```
Blocking phishing email targeting: student@nsbm.ac.lk
```

```
[THREAT LOG] ID: T003 | Source: 203.94.55.12 | Severity:Medium
```

```
!! SECURITY ALERT !! Threat T003 detected from 203.94.55.12
```

```
Mitigating DDoS from 203.94.55.12 -- 5000 packets dropped
```

```
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```